

Name..... Teacher.....

Sydney Technical High School



Mathematics
Preliminary Assessment Task 1
March 2011

General Instructions

- Working Time – 70 minutes.
- Write using a blue or black pen.
- Approved calculators may be used.
- All necessary working should be shown for every question. Marks may be deducted for careless or poorly arranged work.
- Begin each question on a new page.

Total marks (63)

- Attempt Questions 1-8.
- All questions are of equal value.
- Mark values are shown with the questions

Qu 1	Qu 2	Qu 3	Qu 4	Qu 5	Qu 6	Qu 7	Qu 8	Total
								63

Question 1**(8 marks)**

- a) Simplify $\sqrt{75} + \sqrt{32} - \sqrt{27}$ 2
- b) Express $0.\dot{1}2\dot{5}$ as a fraction in simplest form. 1
- c) Evaluate $\frac{\sqrt{16.98+9.074}}{4.99}$ correct to 3 significant figures 2
- d) Simplify $\frac{x^2 + 12x + 36}{x^2 + 6x}$ 2
- e) Fully factorise $8 - 27d^3$ 1

Question 2 (start a new page)**(8 marks)**

- a) Solve for x :
- (i) $\frac{2x}{3} - 4 = x + 2$ 2
- (ii) $(x + 3)^2 = 7$ 2
- b) Express $\frac{x+1}{x^2-1} - \frac{x-1}{x+1}$ as a fraction in its lowest terms 3
- c) Express $a^{-3} \times (8a^6)^{\frac{1}{3}}$ in simplest form, without the use of negative indices 1

Question 3 (start a new page)**(8 marks)**

- a) Solve simultaneously $x - 4y + 12 = 0$ and $y = 3 - 2x$. 3
- b) Solve $2x^2 - 5x - 12 = 0$ 2
- c) Solve and sketch the solution set of $|4 - 3x| < 7$ 3

Question 4 (start a new page)**(8 marks)**

- a) Sketch the function and state the domain and range of the function

3

$$y = \frac{1}{2x - 3}$$

- b) When a number is subtracted from its square the result is 56. Write an equation to represent this information and hence find the answer.

3

- c) Solve for x: $3^x = 9^{x-2}$

2**Question 5 (start a new page)****(8 marks)**

- a) Explain whether $f(x) = x^3 - x$ is an odd function, even function or neither.

2

- b) State the domain and range of the function $y = 2^{-x}$

2

- c) Draw a neat sketch of $y = x^2 - 7x + 12$ showing x and y intercepts and vertex
Hence state the domain and range.

4**Question 6 (start a new page)****(8 marks)**

- a) Simplify $\frac{m^3 + m^2}{x^2 - x} \div \frac{m+1}{x - x^3}$ as a single fraction in simplest form.

3

- b) Fully factorise $w^4 - 16$

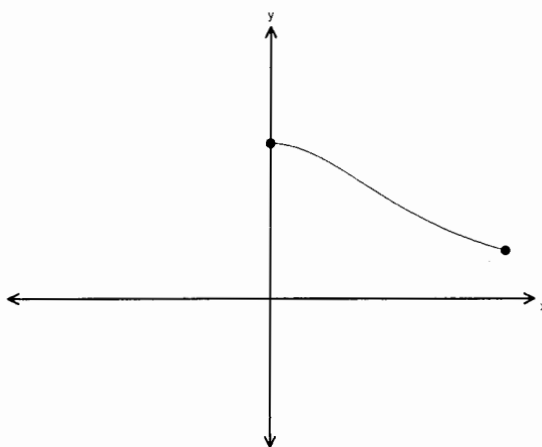
2

- c) Show that $\frac{3\sqrt{2} - 2\sqrt{3}}{3\sqrt{2} + 2\sqrt{3}}$ can be expressed in the form $a + b\sqrt{6}$ and find a and b .

3

Question 7 (start a new page)**(7 marks)**

- a) Solve $x^2 - 3x - 1 = 0$ correct to 2 decimal places. 2
- b) Solve $|2x - 2| = 6x + 10$ Check your solution 3
- c) i) What is the condition for an even function 1
ii) Complete the graph of the function on your answer sheet, so it represents an even function 1

**Question 8 (start a new page)****(8 marks)**

- a) If $f(x) = 3x^2 - 5x + 4$ and $g(x) = 2x + 10$ find:
- i) $f(2) + f(-2)$ 1
- ii) the values of x for which $f(x) = 6$ 2
- iii) the values of x for which $f(x) = g(x)$ 2
- b) Show the region of the number plane where the following hold simultaneously: 3
- $$(x - 2)^2 + y^2 \leq 4$$
- $$y \leq 0$$

END OF EXAMINATION



QUESTION 1

$$a) \sqrt{75} + \sqrt{32} - \sqrt{27} = 5\sqrt{3} + 4\sqrt{2} - 3\sqrt{3} \\ = \underline{\underline{2\sqrt{3} + 4\sqrt{2}}}$$

$$b) \text{ Let } x = 0.\dot{1}\dot{2}\dot{5} \\ 10x = 1.\dot{2}\dot{5} \\ 1000x = 125.\dot{2}\dot{5} \\ 990x = 124 \\ x = \frac{124}{990} \\ \therefore 0.\dot{1}\dot{2}\dot{5} = \underline{\underline{\frac{62}{495}}}$$

$$c) \underline{\underline{1.02}} \text{ (3 sig fig)}$$

$$d) \frac{(x+6)(x+6)}{x(x+6)} = \underline{\underline{\frac{x+6}{x}}}$$

$$e) 2^3 - (3d)^3 = \underline{\underline{(2-3d)(4+6d+9d^2)}}$$

Question 2

$$a) i) \frac{2x}{3} - 4 = x + 2 \\ 2x - 12 = 3x + 6 \\ -18 = x$$

$$\therefore x = \underline{\underline{-18}}$$

$$ii) \underline{\underline{(x-11)(x+8)}}$$

$$b) \frac{(x+1)}{(x-1)(x+1)} - \frac{(x-1)}{x(x+1)} \\ = \frac{x(x+1) - (x-1)^2}{x(x-1)(x+1)}$$

$$\frac{x^2 + x - [x^2 - 2x + 1]}{x(x-1)(x+1)}$$

$$\frac{\cancel{x^2} + x - \cancel{x^2} + 2x - 1}{x(x-1)(x+1)} \\ = \underline{\underline{\frac{3x-1}{x(x-1)(x+1)}}}$$

$$c) \frac{x+1}{(x+1)(x-1)} - \frac{x-1}{x+1} = \frac{(x+1)^2 - (x-1)^2}{(x+1)(x-1)} \\ = \underline{\underline{\frac{4x}{(x+1)(x-1)}}}$$

$$d) a^{-3} \times 2a^2 = 2a^{-1} \\ = \underline{\underline{\frac{2}{a}}}$$

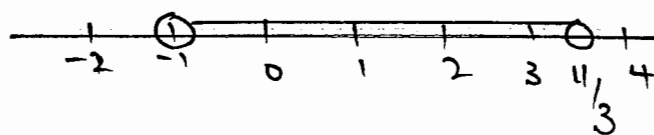
Question 3

$$a) x - 4(3 - 2x) + 12 = 0 \\ x - 12 + 8x + 12 = 0 \\ 9x = 0 \\ \therefore x = 0$$

$$\underline{\underline{y = 3}}$$

$$b) x = \frac{5 \pm \sqrt{25 - 4 \cdot 2 \cdot -12}}{4} \\ x = \frac{5 \pm \sqrt{121}}{4} \\ x = \underline{\underline{4, -3/2}}$$

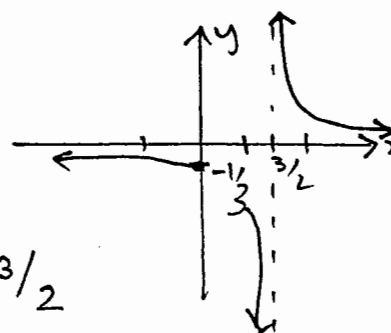
$$c) 4 - 3x < 7 \quad 4 - 3x > -7 \\ -3x < 3 \quad -3x > -11 \\ x > -1 \quad x < 11/3 \\ -1 < x < 11/3$$



Question 4

$$a) 2x - 3 = 0 \\ 2x = 3 \\ \therefore x = 3/2 \\ \therefore D: x \neq 3/2$$

$$R: \text{all } y, y \neq 0$$

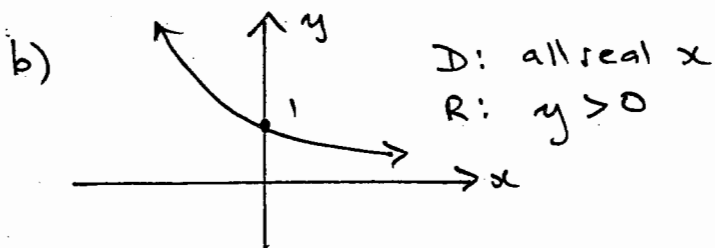


a) let x be number
 $x^2 - x = 56$
 $x^2 - x - 56 = 0$
 $(x+7)(x-8) = 0$
 $x = -7, x = 8$

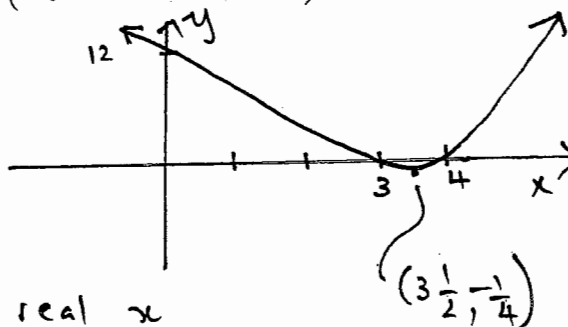
b) $3^x = 3^{2(x-2)}$
 $x = 2x - 4$
 $\therefore 4 = x$
ie $x = 4$

Question 5

a) $f(x) = x^3 - x$
 $f(-x) = (-x)^3 + x$
 $= -x^3 + x$
 $-f(-x) = x^3 - x$
 $\therefore f(x) = -f(-x)$ odd



c) $y = x^2 - 7x + 12$
 $y = (x-3)(x-4)$



D: all real x
R: $y \geq -4$

Question 6

a) $\frac{m^2(m+1)}{x(x-1)} \times \frac{x(1-x^2)}{(m+1)}$

$$\frac{m^2(1-x)(1+x)}{-(1-x)}$$

$$\underline{\underline{-m^2(1+x)}}$$

b)

$$\frac{w^4 - 16}{(w^2 - 4)(w^2 + 4)} = \frac{(w-2)(w+2)(w^2 + 4)}{(w-2)(w+2)(w^2 + 4)}$$

c) $\frac{3\sqrt{2} - 2\sqrt{3}}{3\sqrt{2} + 2\sqrt{3}} \times \frac{3\sqrt{2} - 2\sqrt{3}}{3\sqrt{2} - 2\sqrt{3}}$

$$\frac{18 - 6\sqrt{6} - 6\sqrt{6} + 12}{6} = \frac{30 - 12\sqrt{6}}{6}$$

$$5 - 2\sqrt{6}$$

$\therefore a = 5, b = -2$

Question 7

a) $x = \frac{3 \pm \sqrt{9 - 4 \cdot 1 \cdot 1}}{2}$
 $x = \frac{3 \pm \sqrt{5}}{2}$

$x = 3.30, -0.30$

$$\begin{aligned} \text{b)} \quad 2x - 2 &= 6x + 10 \\ -12 &= 4x \\ x &= -3 \end{aligned}$$

check:

$$\begin{aligned} |-6-2| &= -18+10 \\ 8 &\neq -8 \end{aligned}$$

$\therefore x = -3$ not solution

$$2x - 2 = -6x - 10$$

$$8x = -8$$

$$x = -1$$

check:

$$|-2-2| = -6+10$$

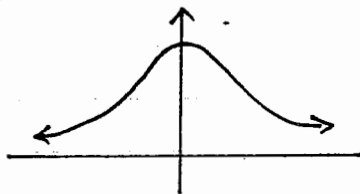
$$4 = 4$$

true

$\therefore \underline{\underline{x = -1 \text{ only solution}}}$

$$\text{c) i) } f(x) = f(-x)$$

ii)



Question 8

$$\begin{aligned} \text{a) i) } f(2) + f(-2) &= (12 - 10 + 4) + (12 + 10 + 4) \\ &= \underline{\underline{28}} \end{aligned}$$

$$\begin{aligned} \text{ii) } 3x^2 - 5x + 4 &= 6 \\ 3x^2 - 5x - 2 &= 0 \\ (3x+1)(x-2) &= 0 \\ \underline{\underline{x = -1/3, \quad x = 2}} \end{aligned}$$

$$\begin{aligned} \text{iii) } 3x^2 - 5x + 4 &= 2x + 10 \\ 3x^2 - 7x - 6 &= 0 \\ (3x+2)(x-3) &= 0 \\ \underline{\underline{x = -2/3, \quad x = 3}} \end{aligned}$$

b)

